1. Mix: real + synthetic: design and factors

Research question: What choices does the experiment vary?

Method: Analyze this training dataset independently. The factorial grid is 5 datasets x 2 generators x 2 has_y values x 3 labelers x 2 seeds = 120 macro-F1 scores.

Conclusion: All relabeled combinations are observed. The 20 generated-target runs and 10 real-only classification runs remain reference comparisons, outside this factorial ANOVA.

Factor	Meaning	Levels
dataset	Real-data prediction task.	Adult; Census KDD; Covertype; MNIST12; MNIST28.
generator	Model fitted on real training data to generate synthetic rows.	CTGAN; TVAE.
has_y	Whether the generator is fitted with target Y.	0: fit X only. 1: fit (X,Y).
labeler	Predictor fitted on real training data to replace synthetic targets.	RF: random forest. XGB: XGBoost. DNN: neural network.
seed	Categorical experiment random-number setting.	42; 43. This is not a continuous covariate.

Reference	Construction	Role
Generated targets	Fit (X,Y), retain generated Y. No labeler.	Two generators x two seeds x five tasks = 20 scores.
Real only	Train downstream model on original real training rows.	Two seeds x five tasks = 10 scores.

Training rows: all real training rows plus 100000 synthetic rows. Mix concatenates all rows; it is not a fixed 50/50 mixture. Both reports use the same experimental scope, with separate statistical fits.

X = features; **Y** = target. has_y describes generator fitting, not whether the final training table contains Y. Every relabeled table contains X and predicted Y. Generated targets have no has_y=0 case and no labeler; that nonexistent combination is never estimated.

Macro F1 is the equal-weight average of F1 across target classes, on a 0-1 scale. The downstream evaluator is a dataset-specific neural model. RF/XGB/DNN are labelers. News and Housing regression scores are analyzed separately on page 5.

2. ANOVA: which factors affect macro F1?

Research question: Do generator, has_y, labeler, seed or their interactions change relabeled-data utility?

Method: Type II ANOVA: macro_F1 ~ dataset + generator * has_y * labeler * seed. All five factors are categorical; sum-to-zero coding is used.

Conclusion: Labeler differs (nominal p = 7.85e-06). Generator and has_y have small average effects; seed and treatment interactions are nonsignificant in this model.

```
macro_F1 ~ dataset + generator * has_y * labeler * seed
```

Term	SS	df	F	Nominal p	Sig.
dataset	**1.8170**	**4**	**302.08**	**5.36e-52**	***
generator	4.27e-09	1	2.84e-06	0.9987	
has_y	0.0001	1	0.04	0.8459	
labeler	**0.0403**	**2**	**13.39**	**7.85e-06**	***
seed	4.81e-05	1	0.03	0.8584	
generator x has_y	0.0001	1	0.05	0.8276	
generator x labeler	0.0001	2	0.02	0.9786	
has_y x labeler	0.0002	2	0.07	0.9292	
generator x seed	3.55e-05	1	0.02	0.8782	
has_y x seed	0.0002	1	0.16	0.6943	
labeler x seed	0.0002	2	0.06	0.9417	
generator x has_y x labeler	0.0002	2	0.08	0.9215	
generator x has_y x seed	1.05e-05	1	0.01	0.9336	
generator x labeler x seed	0.0004	2	0.14	0.8659	
has_y x labeler x seed	0.0007	2	0.23	0.7977	
generator x has_y x labeler x seed	0.0002	2	0.05	0.9495	
Residual	0.1383	92			

Symbols: *** p < .001; ** p < .01; * p < .05; . p < .10. Bold: p < .05. Uses nominal Type II ANOVA p.

* includes all main effects, two-way, three-way and four-way interactions among generator, has_y, labeler and seed. For example, has_y x labeler asks whether including Y helps RF, XGB and DNN differently. Dataset is an additive block in this equation.

Marginal comparison	Mean F1 change	95% model CI
TVAE - CTGAN	+1.19e-05	[-0.0140, +0.0141]
has_y=1 - has_y=0	+0.0014	[-0.0127, +0.0154]
Seed 43 - seed 42	+0.0013	[-0.0128, +0.0153]

SS = sum of squares; df = numerator degrees of freedom; F = term mean square / residual mean square. Residual df = 92. Intervals use the ordinary model assumptions and are unadjusted. Shared test partitions make nominal ANOVA inference exploratory.

3. Does replacing generated targets improve utility?

Research question: Do RF/XGB/DNN targets outperform the generator's own targets, and does has_y matter?

Method: Match within dataset, seed and generator. delta[d] = mean(s,g,labeler) F1_relabel[d] - mean(s,g) F1_generated[d]. Test five dataset differences with t = mean(delta)/(SD(delta)/sqrt(5)), df=4.

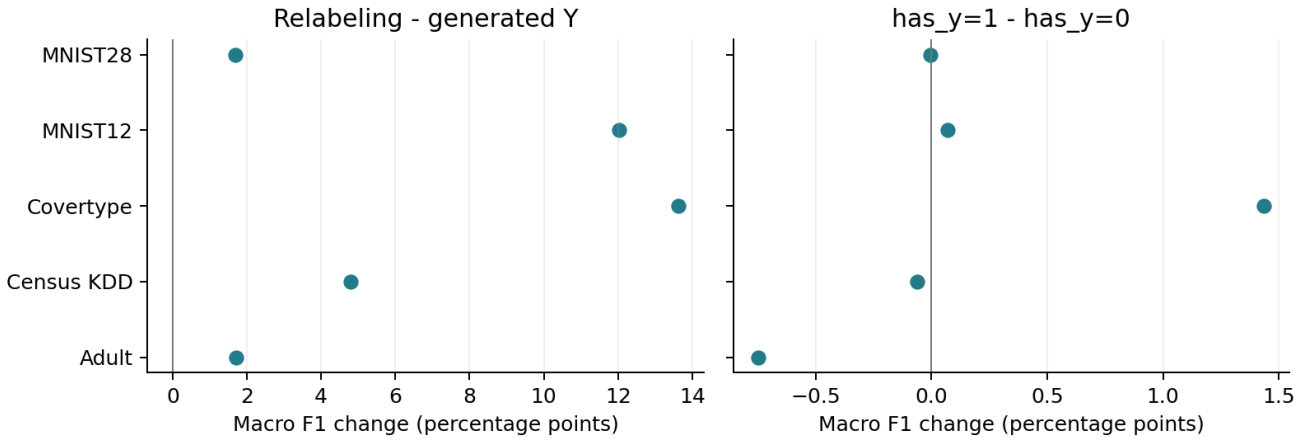
Conclusion: Relabeling gains **+0.0677 macro F1 (9.04% relative)**. has_y=1 minus 0 is **+0.0014**. Neither main comparison passes Holm-adjusted p < .05 across tasks.

Comparison	First mean	Second mean	F1 change	Relative
Relabeled - generated	0.8166	0.7488	+0.0677	9.04%
has_y=1 - has_y=0	0.8172	0.8159	+0.0014	0.17%
has_y=1 - generated	0.8172	0.7488	+0.0684	9.14%
has_y=0 - generated	0.8159	0.7488	+0.0670	8.95%

Comparison	95% paired-t CI	Holm p	Sig.
Relabeled - generated	[-0.0031, +0.1385]	0.1997	
has_y=1 - has_y=0	[-0.0085, +0.0113]	0.7182	
has_y=1 - generated	[-0.0063, +0.1431]	0.1997	
has_y=0 - generated	[+3.20e-05, +0.1340]	0.1997	

Symbols: *** p < .001; ** p < .01; * p < .05; . p < .10. Bold: p < .05. Uses Holm-adjusted dataset-level p over four planned contrasts.

Why the gain misses Holm significance: the five task gains range from 1.70 to 13.63 percentage points; SE = 2.55 points. Raw p = 0.0566; Holm p = 0.1997. The smallest of four raw p-values is 0.0499, above the first Holm cutoff .05/4 = .0125. None passes. Large gains do not imply precise estimates.



Five dataset means show the main contrasts. Positive values favor relabeling or has_y=1.

All datasets, seeds and generators receive equal weight; relabeling also averages the three labelers. has_y compares only relabeled runs. The generated baseline is matched once per dataset/seed/generator and is not treated as six independent copies.

Relative gain = F1 change / second mean; .01 F1 is one percentage point. Confidence intervals are unadjusted. Exact sign-flip results are supplied; with five tasks the minimum two-sided p is .0625. This is not evidence that every future task improves.

4. How large is each labeler's improvement?

Research question: For each generator and has_y, how much do RF, XGB and DNN improve on generated targets?

Method: delta[d] = mean(s) F1_relabel[d,g,has_y,labeler] - mean(s) F1_generated[d,g]. Paired t-test across five datasets, df=4; Holm over twelve comparisons.

Conclusion: Largest descriptive gain: CTGAN / DNN / has_y=1 (**+0.1277**). 0 of twelve baseline comparisons pass Holm-adjusted p < .05.

Generator	Labeler	has_y	F1 mean	F1 gain	Relative	Holm p	Sig.
CTGAN	RF	0	0.8023	+0.0857	11.97%	0.9319	
CTGAN	XGB	0	0.8068	+0.0902	12.59%	0.9319	
CTGAN	DNN	0	0.8408	+0.1243	17.34%	0.8734	
CTGAN	RF	1	0.7958	+0.0793	11.06%	0.9319	
CTGAN	XGB	1	0.8094	+0.0928	12.96%	0.9319	
CTGAN	DNN	1	0.8442	+0.1277	17.82%	0.8782	
TVAE	RF	0	0.7971	+0.0160	2.05%	0.9319	
TVAE	XGB	0	0.8066	+0.0255	3.26%	0.9319	
TVAE	DNN	0	0.8416	+0.0605	7.75%	0.9319	
TVAE	RF	1	0.8000	+0.0189	2.42%	0.9319	
TVAE	XGB	1	0.8136	+0.0325	4.16%	0.9319	
TVAE	DNN	1	0.8405	+0.0594	7.60%	0.9319	

Symbols: *** p < .001; ** p < .01; * p < .05; . p < .10. Bold: p < .05. Uses Holm-adjusted dataset-level p across twelve baseline comparisons.

Matched generated-target comparator	Mean macro F1
CTGAN	0.7165
TVAE	0.7811

These are comparisons against each generator's own generated targets, not pairwise tests between labelers. The labeler ANOVA on page 2 tests differences among RF/XGB/DNN. An observed largest gain does not establish a universally best labeler.

Means average seeds within tasks and then give each task equal weight. Per-task differences, confidence intervals, exact tests and fitted marginal means are in the supplied tables. Generated baselines remain outside the factorial fit.

5. Regression: what changes on News and Housing?

Research question: Does relabeling improve regression utility under the same constructions?

Method: Compare saved scores with generated targets. Means average seeds, generators and, for relabeled runs, RF/XGB/DNN. Pages 7-8 separately test factor effects within relabeled runs.

Conclusion: Housing relabeling improves all three average metrics. News relabeling improves R2 but worsens NMAE and D2. Baseline gains below are descriptive; ANOVA tests are exploratory.

```
R2 = 1 - sum((Y - prediction)^2) / sum((Y - mean(Y))^2)
```

```
NMAE = mean(abs(Y - prediction)) / SD_test(Y), ddof = 0
```

```
D2 = 1 - MAE_prediction / MAE_test_median_baseline
```

Dataset / metric	Real only	Generated Y	Relabeled has_y=0	Relabeled has_y=1
News / R2	-0.0290	-0.0196	0.0002	-0.0008
News / NMAE	0.2686	0.2494	0.2551	0.2544
News / D2	-0.0445	0.0300	0.0079	0.0104
Housing / R2	0.5125	0.6676	0.7905	0.7872
Housing / NMAE	0.5507	0.4137	0.3103	0.3052
Housing / D2	0.2822	0.4607	0.5956	0.6022

Relabeled - generated	R2 change	D2 change
News	+0.0192	-0.0208
Housing	+0.1212	+0.1382

Higher R2/D2 is better; lower NMAE is better. R2 measures squared error against the test-mean baseline; D2 measures absolute error against the test-median baseline. Negative R2/D2 means worse than that baseline.

News uses log-target reruns scored in original units; Housing uses no-Faker reruns. NMAE and D2 read saved test normalization and verified sidecars; denominators are not recomputed for this report. Real-only runs are references.

Scope: 406 verified selected records across both training analyses: 290 classification and 116 regression, plus 116 verified D2 sidecars. The 14 real-only runs are shared references. Credit, Intrusion, NB/PCA-GMM and simulated experiments are excluded.

MNIST caveat: output heads were repaired; retained MNIST12 TVAE synthetic tables still have documented holdout-feature collisions. Seeds share prepared test partitions. Neither seed supplies an independent new test sample.

6. Model checks: do common effects describe each task?

Research question: Do classification effects vary by task, and which regression factors matter within News or Housing?

Method: Use nested F tests for the classification equations below. For each regression dataset, fit R2 or D2 ~ seed + generator * has_y * labeler.

Conclusion: Classification dataset interactions are strong (p = 2.93e-32). Regression effects also depend on the task; use the within-task p-values below and signed changes on pages 7-8.

`Seed block: macro_F1 ~ dataset + seed + generator * has_y * labeler`

`Dataset check: macro_F1 ~ dataset * (has_y * labeler)`

`+ generator * has_y * labeler + seed`

Added terms	df num / den	F	Nominal p	Sig.
Seed interactions	11 / 92	0.105	0.9999	
Dataset interactions	**20 / 83**	**35.220**	**2.93e-32**	***

Symbols: *** p < .001; ** p < .01; * p < .05; . p < .10. Bold: p < .05. Uses nominal nested-model p; these are supporting checks.

`Within each regression dataset: R2 or D2 ~ seed + generator * has_y * labeler`

Dataset / outcome	Generator p	has_y p	Labeler p	Seed p
News / R2	0.5895	0.2084	0.1594	0.9051
Housing / R2	**0.0037 ****	0.4679	**0.0066 ****	0.5729
News / D2	0.3946	0.4105	0.0856 .	0.1115
Housing / D2	**8.22e-05 *****	**0.0140 ***	**6.35e-08 *****	0.9061

Symbols: *** p < .001; ** p < .01; * p < .05; . p < .10. Bold: p < .05. Uses nominal within-task ANOVA p; bold cells mark p < .05.

Each regression task has 24 relabeled scores, 13 coefficients and 11 residual df. A full generator * has_y * labeler * seed fit within one task would be saturated: 24 coefficients, zero residual df. The stated within-task equation keeps seed additive; its omitted seed interactions form the residual. Full tables are supplied.

Primary model: 120 scores, 28 identifiable coefficients, 92 residual df. Seed-block rank = 17; dataset-check rank = 37. All ANOVA F values were independently reproduced. Primary residual Shapiro p = 3.30e-07; adjusted R-squared = 0.9104.

The ordinary ANOVA assumes independent errors with common variance. Shared test sets, nonnormal residuals and dataset-specific treatment effects limit its p-values; page 3 therefore reports paired evidence across five dataset means. Seed is fixed at 42/43, not a sampled population of seeds.

Provenance: original score snapshot Oct 10, 2026, 1:08:42 a.m. Chicago; all 522 source/sidecar hashes rechecked Oct 10, 2026 at 02:53 PM Chicago. Generator settings remain 500 epochs and 100,000 rows. Census selects checkpoints by weighted macro-F1; others use development loss. Experiments were not rerun.

7. Regression ANOVA: which factors affect R2?

Research question: Among relabeled runs, do generator, has_y, labeler, seed or their interactions change R2?

Method: Type II ANOVA: R2 ~ dataset + generator * has_y * labeler * seed. All factors are categorical with sum-to-zero coding; synthetic and mix are fitted independently.

Conclusion: Nominal main effects: generator (p = 0.0423), labeler (p = 0.0499). has_y has no detected common effect; a significant term means choices differ, not that every choice improves utility.

```
R2 ~ dataset + generator * has_y * labeler * seed
```

Term	SS	df	F	Nominal p	Partial eta2	Sig.
dataset	**7.4735**	**1**	**45951.90**	**1.81e-39**	**0.999**	*******
generator	**0.0008**	**1**	**4.62**	**0.0423**	**0.167**	*****
has_y	0.0001	1	0.35	0.5589	0.015	
labeler	**0.0011**	**2**	**3.43**	**0.0499**	**0.230**	*****
seed	2.14e-05	1	0.13	0.7202	0.006	
generator x has_y	2.32e-06	1	0.01	0.9060	0.001	
generator x labeler	0.0004	2	1.35	0.2788	0.105	
has_y x labeler	0.0004	2	1.24	0.3086	0.097	
generator x seed	1.09e-05	1	0.07	0.7976	0.003	
has_y x seed	0.0001	1	0.46	0.5058	0.019	
labeler x seed	0.0003	2	1.04	0.3704	0.083	
generator x has_y x labeler	0.0005	2	1.55	0.2333	0.119	
generator x has_y x seed	1.72e-05	1	0.11	0.7480	0.005	
generator x labeler x seed	9.93e-07	2	3.05e-03	0.9970	2.65e-04	
has_y x labeler x seed	0.0003	2	0.88	0.4296	0.071	
generator x has_y x labeler x seed	3.01e-05	2	0.09	0.9120	0.008	
Residual	0.0037	23				

Symbols: *** p < .001; ** p < .01; * p < .05; . p < .10. Bold: p < .05. Uses nominal exploratory ANOVA p; no multiplicity adjustment.

Marginal comparison	Pooled change	News change	Housing change
TVAE - CTGAN	-0.0079	+0.0004	-0.0163
has_y=1 - has_y=0	-0.0022	-0.0010	-0.0033
Seed 43 - seed 42	-0.0013	-0.0001	-0.0026
RF - DNN	-0.0098	-0.0020	-0.0177
XGB - DNN	+0.0008	-0.0010	+0.0026

Positive changes favor the first choice. Pooled changes give the two tasks equal weight. Within-task model p-values are on page 6; unadjusted 95% model intervals and contrast p-values are supplied in regression_factor_effects.csv.

48 observed scores; 25 identifiable coefficients; 23 residual df. Partial eta2 = SS_term / (SS_term + SS_residual). These exploratory tests use two fixed datasets and shared test partitions. They do not test relabeled versus generated targets; those baseline changes are on page 5. Dataset SS reflects different task score levels.

8. Regression ANOVA: which factors affect D2?

Research question: Among relabeled runs, do generator, has_y, labeler, seed or their interactions change D2?

Method: Type II ANOVA: D2 ~ dataset + generator * has_y * labeler * seed. All factors are categorical with sum-to-zero coding; synthetic and mix are fitted independently.

Conclusion: Nominal main effects: labeler (p = 0.0005). has_y has no detected common effect; a significant term means choices differ, not that every choice improves utility.

```
D2 ~ dataset + generator * has_y * labeler * seed
```

Term	SS	df	F	Nominal p	Partial eta2	Sig.
dataset	**4.1732**	**1**	**22153.69**	**7.92e-36**	**0.999**	***
generator	0.0008	1	4.27	0.0503	0.157	.
has_y	0.0003	1	1.33	0.2605	0.055	
labeler	**0.0040**	**2**	**10.66**	**0.0005**	**0.481**	***
seed	0.0001	1	0.45	0.5071	0.019	
generator x has_y	2.81e-05	1	0.15	0.7030	0.006	
generator x labeler	0.0004	2	1.11	0.3479	0.088	
has_y x labeler	0.0001	2	0.21	0.8090	0.018	
generator x seed	4.68e-05	1	0.25	0.6229	0.011	
has_y x seed	0.0001	1	0.45	0.5090	0.019	
labeler x seed	0.0001	2	0.16	0.8501	0.014	
generator x has_y x labeler	4.73e-05	2	0.13	0.8825	0.011	
generator x has_y x seed	0.0001	1	0.47	0.5012	0.020	
generator x labeler x seed	1.47e-05	2	0.04	0.9618	0.003	
has_y x labeler x seed	2.13e-05	2	0.06	0.9453	0.005	
generator x has_y x labeler x seed	1.69e-05	2	0.04	0.9561	0.004	
Residual	0.0043	23				

Symbols: *** p < .001; ** p < .01; * p < .05; . p < .10. Bold: p < .05. Uses nominal exploratory ANOVA p; no multiplicity adjustment.

Marginal comparison	Pooled change	News change	Housing change
TVAE - CTGAN	-0.0082	-0.0026	-0.0138
has_y=1 - has_y=0	+0.0046	+0.0025	+0.0066
Seed 43 - seed 42	+0.0027	+0.0051	+0.0003
RF - DNN	-0.0190	-0.0081	-0.0299
XGB - DNN	+0.0008	-0.0073	+0.0089

Positive changes favor the first choice. Pooled changes give the two tasks equal weight. Within-task model p-values are on page 6; unadjusted 95% model intervals and contrast p-values are supplied in regression_factor_effects.csv.

48 observed scores; 25 identifiable coefficients; 23 residual df. Partial eta2 = SS_term / (SS_term + SS_residual). These exploratory tests use two fixed datasets and shared test partitions. They do not test relabeled versus generated targets; those baseline changes are on page 5. Dataset SS reflects different task score levels.